EOB/CAT Protocol for CasX Cleavage Assay

Annealing Duplex DNA:

Stock of labeled target strand DNA is 100uL of 100nM.

Mix 1:1.2 molar ratio of ^{32}P labeled target strand to unlabeled nontarget strand of duplex substrate.

Stock of unlabeled TS: for 50uL, 120nM (4.48uL TS stock, 45.42uL DEPC)
Stock of unlabeled NTS: for 50uL, 120nM (3.56uL TS stock, 46.44uL DEPC

To make a 100uL stock of duplex substrate, mix 50uL of 100nM labeled target with 50uL of 120nM unlabeled nontarget strand. (Or reverse concentrations for labeled nontarget, unlabeled target) Place on 95°C heat block for 2-5 minutes, then slow cool for ~2h. This is the **Duplex DNA stock** for cleavage assays. DNA stock was diluted to 120nM with CasX reaction buffer.

CasX Activity Assays:

Dilute CasX into CasX dilution buffer (500 mM NaCl, 10% glycerol, 20mM Tris-HCl, pH 7.5, 1 mM magnesium chloride, 0.5 mM TCEP) to 4uM. Dilute sgRNA into CasX reaction buffer (20 mM HEPES, pH 7.5, 10 mM magnesium chloride, 150mM potassium chloride, 1% glycerol, 0.5 mM TCEP) to 6uM. Follow Excel spreadsheet for stock dilutions of each sample. Mix equimolar ratio of CasX and sgRNA (2uL sgRNA stock, 3uL CasX stock, 1uL reaction buffer + TCEP). Allow samples to mix for 30 minutes at room temperature.

Dilute annealed, labeled duplex DNA into CasX reaction buffer. (2uL stock, 43uL of reaction buffer + TCEP)

Add RNP stock (5uL, 2uM RNP) to the DNA/buffer mixture. Total volume should be 50uL. Incubate at 37°C, and distribute aliquots (5 uL) into tubes filled with quencher (5 uL) for each time point (0,1,5,10,30,60,120 min). Incubate with = volume at RT for 5m with 50ug/mL heparin +EDTA, add 5uL (1/2 total previous volume) formamide loading dye and incubate at 95°C for 5m.

Add 5uL quencher [(50ug/mL heparin, 25mM EDTA] quencher, then add 5uL of formamide loading dye, then 95°C for 5m. **= volume is generally 5uL.

After quenching, samples are loaded on a 12% PAGE gel and run at 40-45 W for 45 min. (After large gel was pre-warmed for ~45 min at 25 W.) Approximately 200 cpm/well was loaded per sample (4uL/well). (Loading less- 100, 50 cpm- works fine. You may need to expose overnight.)